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Original Research Article

Analyzing soil erosion trends and future predictions using the RUSLE model: A case study of the Dwarakeswar River Basin, West Bengal

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ABSTRACT

Climate change and non-sustainable land use management are closely linked to soil erosion (SE), which poses the most significant threat worldwide. This paper aims to assess annual SE in Dwarakeswar River Basin (DRB) in terms of current and future climatic and land-use change in 2035 and 2050. The Revised Universal Soil Loss Equation (RUSLE) model was applied to obtain SE. Furthermore, land use has been simulated using an ANN and Markov chain model. Precipitation was projected using the General Circulation Model (GCM) of MIROC6 (Model for Interdisciplinary Climate Research) from CMIP6 (Coupled Model Intercomparison Project Phase 6) under three Representative Concentration Pathway (RCP) 2.6, 4.5, and 8.5 scenarios. Finally, this study examines the variations between forecasted and current erosion rates to identify the patterns of soil erosion change and the SE pockets. The average rainfall rate in future is expected to be lower than in the current period. Results revealed that the mean average estimated annual SE rate in the study area in 2022 approximately 7.74 t ha^{-1} yr^{-1}. In contrast to RCP 2.6, 4.5, and 8.5 scenarios in 2035 the mean SE 23.56 t $ha^{-1}yr^{-1}$, 22.95 t $ha^{-1}yr^{-1}$, and 22.84 t $ha^{-1}yr^{-1}$, and in 2050 the mean SE 25.16 t $ha^{-1}yr^{-1}$, 24.76 t $ha^{-1}yr^{-1}$, and 25.51 t $ha^{-1}yr^{-1}$ respectively, which represents a 3 to 4 times SE increase in the future. To mitigate this damage, a better management plan should be adopted, which could possibly address this issue, especially in those SE pockets.

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1. Introduction

Water-induced soil erosion (SE) is a noticeable form of land degradation, arising from both natural and anthropogenic activities (Bhattacharya et al., 2020; Rosas & Gutierrez, 2020). This leads to declining ecosystems, biodiversity, and human sustainability globally (Pimentel, 2006; Haregeweyn et al., 2015). Due to unsustainable farming, overgrazing, deforestation, climate change, drought, urbanization, and inadequate land management (Arabameri, Tiefenbacher, Blaschke, Pradhan, & Bui, 2020; Chuenchum et al., 2020a; Hosseinalizadeh et al., 2020; Quesada-Román & Mata-Cambronero, 2021; Quesada-Román & Campos-Durán, 2023), SE is promoted, threatening the economic and ecological sustainability of soil (Visser et al., 2019; Baranian Kabir et al., 2020; Singh, 2023; Hateffardet al., 2021) (see Table 1).

SE is intensifying as a result of unsustainable agricultural practices (Robinson et al., 2017; Chuenchum et al., 2020a), deforestation, and pasturage (Chuenchum et al., 2020b; Quesada-Román & Mata-Cambronero, 2021; Quesada-Román & Campos-Durán, 2023). These factors contribute to both direct and indirect impacts on SE, water quality, the hydrological cycle, its associated systems, and infrastructure (Pimentel, 2006; Bogale et al., 2020). Globally, approximately 75 billion tons of soil are eroded annually, primarily

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from agricultural land, resulting in the loss of about 20 million hectares (Mha) of land (Arabameri et al., 2020; Shit et al., 2015).

Regions such as Asia, Africa (Borrelli et al., 2020; GSP, 2017), and South America experience severe SE, approximately 30–40 t ha^{-1} yr^{-1} (Pimentel, 2006). Whereas in Asia, the typical rate of SE is 138 t ha^{-1} yr^{-1} (Chen et al., 2010; Sharma, 2010; Shit et al., 2015). The average SE from cropland ranges from 0.5 to 400 t ha^{-1} yr^{-1} worldwide (Fayas et al., 2019; Pimentel et al., 1995; Singh & Panda, 2017). In India, approximately 175 Mha of land is adversely affected by SE and also by the different types of land degradation processes, whereas near about 150 Mha of land are affected by active water and wind erosion of subsoil annually, and around 25 Mha of land is reported to have been degraded by the ravine, gullies, slash-and-burn cultivation, and flooding (Reddy, 1999). Narayana and Babu (1983) estimated that India loses approximately 5334 m (i.e., 16.4 t ha^{-1} yr^{-1}) of soil annually, with about 29 % transported to the sea by rivers and only 10 % deposited in reservoirs, resulting in a significant reduction in storage strength. In West Bengal, 14 % of the land area is affected by SE. The rate of SE is high in Purulia district, followed by West Medinipur, Bankura, Koch Bihar, and Jalpaiguri (Pandey et al., 2011). Several studies indicate a decline in crop production in the western part of West Bengal due to SE over the last few decades (Mondal, 2012; Nandi, 2012; Shit et al., 2015). Additionally, SE degrades agricultural productivity and increases water consumption, leading to global economic losses of $8 billion and approximately $1500 million annually in India (Sartori et al., 2019).

Soil is a crucial natural resource, essential for plant growth, healthy ecosystems (i.e., nutrient cycling, water filtration, and carbon storage), and sustaining agriculture, forestry, and biodiversity. However, SE, driven by both natural causes and anthropogenic activity, is gravely degraded. This degradation impacts agricultural productivity, increases river sedimentation and water pollution, destroys habitats, and disrupts ecological balance. To combat this, we must identify micro-level SE zones for effective land and water management to maintain sustainability (Biddoccu et al., 2020). Additionally, predicting future SE helps us anticipate and protect areas where erosion is likely to be intensified.

To measure the SE, various models play a crucial role in estimating and predicting on both a global scale (Biddoccu et al., 2020) and a regional scale. There are different soil erosion models regarding the assessment of SE, such as the Revised Universal Soil Loss Equation (RUSLE), the Erosion Productivity Impact Calculator (EPIC) (Williams, Jones, & Dyke, 1990), the European Soil Erosion Model (EUROSEM) (Morgan et al., 1998), and the Water Erosion Prediction Project (WEPP) . Recently, advanced techniques such as Artificial Neural Networks (ANN) and Machine Learning (ML) have gained popularity for erosion simulation and modelling (Avand et al., 2023; Gholami et al., 2018; Pourghasemi, Sadhasivam, Kariminejad, & Collins, 2020; Roy et al., 2023). Among those models, the Universal Soil Loss Equation (USLE) (Wischmeier & Smith, 1978) and the Revised Universal Soil Loss Equation (RUSLE) (Jaman et al., 2024; Renard, 1997) are the most widely used. So, this model has been employed in this study.

Recent studies exhibits climate-related land degradation is a prominent form of SE (Aslan et al., 2019; Stefanidis & Stathis, 2018). The Intergovernmental Panel on Climate Change (IPCC) has outlined four future scenarios known as Representative Concentration Pathways (RCPs), based on greenhouse gas emissions: 2.6, 4.5, 6.0, and 8.5 (Dash & Maity, 2023; Hateffard et al., 2021; Maurya et al., 2021; Tan et al., 2017). This model is known as a General Circulation Model (GCM), which can predict future climate trends (IPCC, 2013, 2019; Johnson et al., 2011). However, GCMs need to be bias-corrected (Hoang et al., 2016; Mandle et al., 2017). Previous studies have used the Land Change Modeler (LCM) within TerrSet software to manage areas affected by SE. It is highly effective for modelling future Land Use Land Cover (LULC) dynamics (Anand & Oinam, 2020; Khoshnood Motlagh et al., 2021; Zarandian, Mohammadyari, Mirsanjari, & Visockiene, 2023; Gaur & Singh, 2023; Ait El Haj et al., 2023; Mitra et al., 2023). This model uses a neural network to create transition potential maps and employs a Markov Chain to simulate future LULC. Despite numerous studies on SE estimation, there is a lack of information regarding different climatic and topographic patterns in the same basin. Dwarakeswar River Basin (DRB) has its own unique characteristics, with the upper part being semiarid and the middle and lower parts experiencing a monsoon climate. The present study examines how SE differs in different climatic zones and how future LULC and climate changes may influence SE.

The study presents an integration of the RUSLE with the spatio-temporal Remote Sensing and GIS techniques to assess trends of soil erosion in the DRB of West Bengal. Despite the extensive global application of the RUSLE model for SE studies, limited research has focused on its implementation in the context of medium-sized river basins in eastern India, particularly the DRB in West Bengal. This region is characterized by a complex mix of topography, rainfall variability, and increasing anthropogenic pressure; yet, it lacks comprehensive, GIS-based soil erosion assessments that integrate both historical trends and future predictions. Unlike previous studies, this study employs spatio-temporal satellite imageries to analyze land use land cover (LULC) changes over time and projects future erosion scenarios based on present anticipated land transformations. The model is further enhanced by incorporating locally calibrated RUSLE parameters, including rainfall erosivity and cover-management factors (C factor), which increase the reliability and accuracy the estimation of soil loss. Validation of the model using historical rainfall data and field observations strengthens the scientific robustness of the findings. Existing studies have either focused on broader national or state-level erosion estimates or lacked spatial precision. Consequently, there is a significant research gap in understanding localized erosion dynamics and forecasting future risks. Therefore, this study aims to address this issue by applying a high-resolution, multi-temporal RUSLE modelling approach supported by satellite data and field validation (Fig. S7).

Prediction of future SE by LULC and climate change scenarios must be anticipated to protect the areas where erosion is likely to be intensified in DRB. The main objectives of this study are to assess soil erosion trends in the past and present times in the DRB using the RUSLE model. Secondly, this study predicts the C factor and P factor in 2035 and 2050 using a future LULC simulation method. This study also identifies the trends of rainfall erosivity (R) factor in

Table 1
Attributes of Dwarekeshwar River.

Extension	latitudes 23°22'27"N -22°39'30"N and longitudes 86°30'56"E–87°47'12"E
Catchment area	4340.8 square kilometers
Origin	Tilaboni hills, near Bangalia railway station
Climate zone	Tropical Region (dominated by Monsoon Climate)
Toposheets no	73 I/10, 73 I/11, 73 I/12, 73 I/14, 73 I/15, 73 I/16, 73 M/3, 73 M/4, 73 M/7, 73 M/8, 73 M/12, 73 M/16, 73 N/9, 73 N/13, and 73 N/14.
Rainfall	117.76 mm/per year
Temperature	32.55 °C

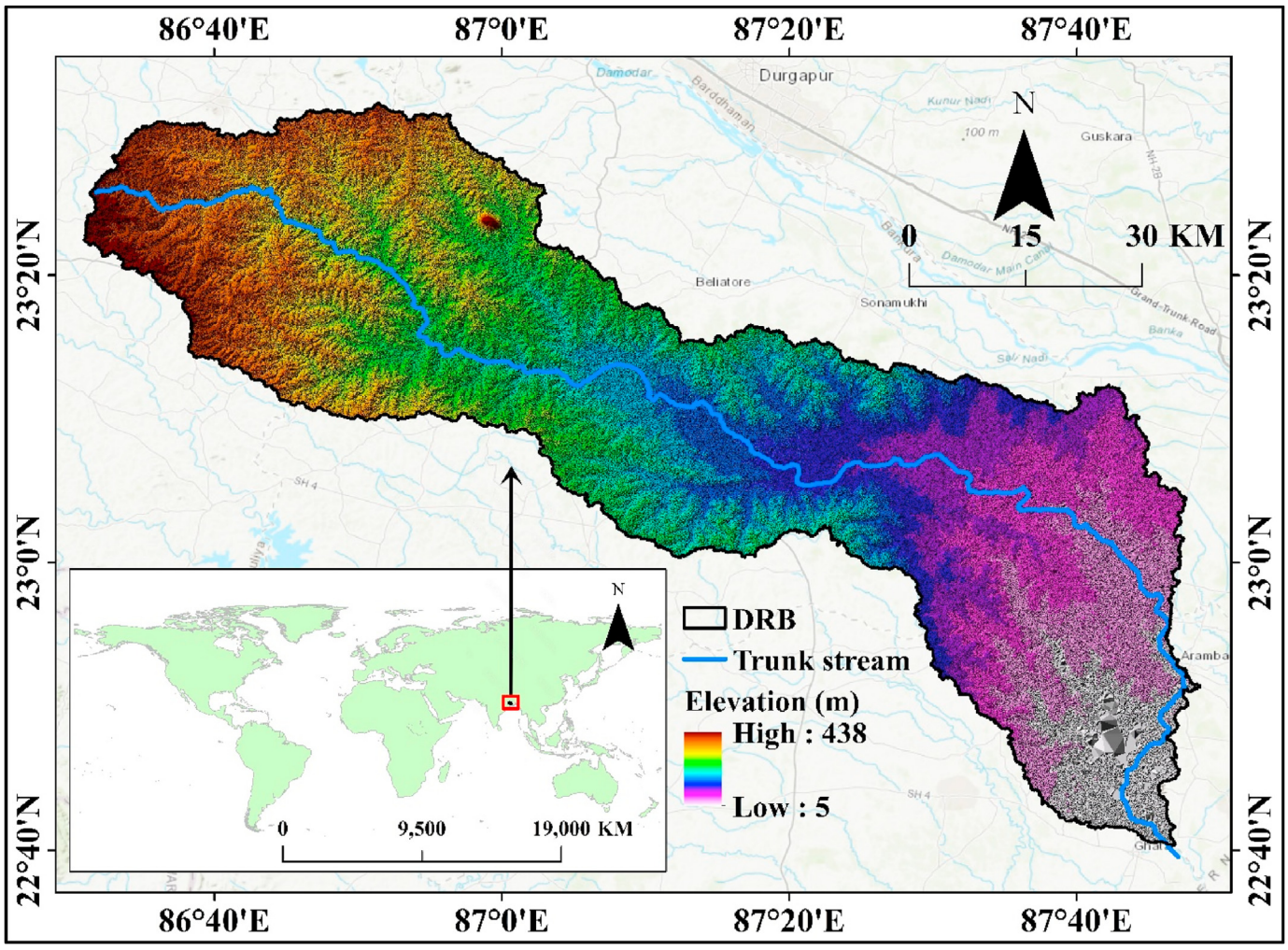


Fig. 1. Location map of the study area.

contrast to GCMs under RCP 2.6, 4.5, and 8.5 scenarios for 2035 and 2050. Finally, this study examines the variations between forecasted and current erosion rates to identify patterns and pockets of SE changes in the DRB for better management practices. The findings aim to provide policymakers and stakeholders with critical information to develop effective strategies for sustainable land resource planning, exploitation, and management.

2. Study area

Fig. 1 represents the Dwarakeswar River basin (DRB), a significant river in western West Bengal, originating in the Tilaboni hills near Bangalia railway station in the Purulia district. This river merges with the Shilabati River, which subsequently joins the Rupnarayan River near Ghatal (Table 1). The watershed encompasses approximately 4340.8 km^2, with a maximum elevation of 438 m (Fig. 4(a)) and a mean slope of 1.71° (Fig. 4(b)). The DR plays a crucial role in the local ecology, culture, and economy of the region. The DRB features a semielliptical shape, spanning the Kashipur block in the northeastern part of Purulia and extending predominantly into the Chhatna block of Bankura. This watershed is characterized by Precambrian crystalline formations and recently deposited alluvium, connected by an intermediate tract (Mukherjee, 1995). This basin is classified under the tropical wet and dry (Aw) climate zone according to Köppen's classification. The basin exhibits varied climatic conditions. The upper part, located in the Plateau province, depicts semiarid environments with an undulating surface and sparse vegetation. In contrast, the middle and lower regions have a tropical monsoon climate, with tropical deciduous forests. The region has experienced an average annual rainfall of approximately 1357 mm (Singha et al., 2025), with more than 80 % of the rainfall occurring during the monsoon season. The annual average temperature is about 32.55 °C, with daily temperature ranges between 4 °C and 9 °C (Sinha, 2016; Kundu & Nag, 2018). The western side of the basin is steeply undulating, whereas the eastern portion is relatively flat. The basin's geology ranges from Archaean granite, quartzite, and schist to Holocene sand and silt deposits (Mukherjee, 1995). Major geological formations include the Pleistocene Lalgarh Formation, the Sijua Formation, and the Holocene Chinsura Formation (Geological Survey of India, 2006). The primary soil types are Typic Haplustalfs and Vertic Haplaguepts. The middle part is characterized by brittle laterite and lateritic soils, which are prone to significant rill and gully erosion, leading to substantial SE.

3. Material and methods

3.1. Data used and pre-processing (Table 2)

3.1.1. Methodology

The RUSLE model has been employed to estimate soil erosion for

Table 2
Different variables of the RUSLE model and data sources.

Factor	Environmental variables	Resolution	Source
Terrain	SRTM DEM	30m	https://earthexplorer.usgs.gov/
	Slope	30m	
Climate	Rainfall data	0.25 x 0.25°	https://www.imdpune.gov.in/cmpg/Griddata/25_NetCDF.html.
	RCP 2.6, 4.5, and 8.5 scenario	30 arcsec (~1 km)	https://chelsa-climate.org/
Vegetation	NDVI	30m	https://earthexplorer.usgs.gov/
Land Cover	Land Use Land Cover type (LULC)	30m	https://earthexplorer.usgs.gov/
Soil	Soil type	250m	https://soilgrids.org/
	Sand		
	Silt		
	Clay		
	OC		

the last 25 years, and these 25 years are further divided into 4 periodical changes: 1998, 2006, 2014, and 2022, which have been plotted in raster format. To accurately assess the impact of climate change on SE in the future, the RCP 2.6, 4.5, and 8.5 scenarios have been used to model mean annual precipitation for the near future (2035 and 2050), and also to anticipate the effects of LULC change, LULC predictions for 2035 and 2050 have been made to simulate the C and P factors in the near future.

Only the R, C, and P factors have units for future prediction of SE, and the remaining parameters are constant. The K factor, which is depends on soil characteristics, is constant due to the slow process of pedogenesis, where it can take centuries to form just 1 cm of new soil under favorable conditions (Bandyopadhyay et al., 2021; Bockheim, 2014; Osman, 2018, pp. 255–298; Van Breemen & Buurman, 2002). K tends to remain constant because residual soils are derived from ancient bedrock, as observed in the study area (Mandal et al., 2018). Similarly, the LS factor, representing slope length and steepness, exhibits relative stability over human timescales. The dynamic equilibrium theory suggests that while landscapes are constantly evolving through erosion and deposition, these processes achieve a balance over extended periods (Hack, 1975). Only significant endogenetic events can disrupt this geomorphic equilibrium and alter the LS factor (Hack, 1975; Wang et al., 2021).

The soil erosion map has been employed by superimposing layers of each RUSLE component in a GIS environment. The RUSLE model has been employed, based on the methodology proposed by Renard (1997) to evaluate routine SE (Eq. (1)).

$$A = R*K*L*S*C*P \quad (1)$$

Where A represents soil loss in tons per hectare per year (t ha^{-1} yr^{-1}), the 'soil erodibility' is defined by K, the 'rainfall erosivity' is denoted by R, the S is 'slope steepness', L represents the 'slope length', the 'cover management' is denoted by C, and P is denoted as the 'conservation practice'. The values for L, S, C, and P are dimensionless. To generate a soil erosion map, all these elements (RKLSCP) are further overlaid using the map algebra tool in ArcGIS environment. The methodology of the proposed soil erosion model is illustrated in Fig. 2. The estimated soil erosion rates were categorized into six levels:

3.2. Rainfall erosivity factor (R)

The rainfall erosivity (R) factor describes the capacity of SE from a bare surface by rainfall intensity (Renard, 1997); it's the mean indelible annual precipitation value from a specific area, relying on the quantity, intensity, velocity, size, and kinetic energy of the rainfall. For this study, rainfall data from the previous 30 years (1993–2022) provided by the IMD Pune (Fig. S1) have been used to calculate the R factor. To calculate the R-factor, the modified Fournier index of Wischmeier and Smith (1978) and later on modified by Arnoldus (1980) has been employed (Eq. (2)). This method is widely accepted for tropical regions to estimate SE of the small and large river basins (Ganasri & Ramesh, 2016; Prasannakumar et al., 2012).

For future rainfall prediction, the CHELSA (Climatologies at high resolution for the earth's land surface areas) database, version 2, has been used and the Modified Fournier index has been employed to calculate future R. CHELSA is a worldwide raster database with ~1 km resolution, including present, future, and past climate data (Karger et al., 2017). The selected climate model (MIROC6) from CMIP6, under three carbon dioxide emission scenarios (RCP 2.6, 4.5, and 8.5), projects average precipitation for two time periods: 2035 and 2050.

$$R = \sum_{i=1}^{12} 1 \cdot 735 \times 10^{[1.5 \log_{10}(P_i^2/P) - 0.08188]} \quad (2)$$

Where rainfall erosivity is represented as R (MJ mm ha^{-1} h^{-1} yr^{-1}), P_i is the monthly rainfall (mm) and P is the annual rainfall (mm).

3.3. Soil erodibility factor (K)

The soil erodibility factor (K) was calculated using texture and organic carbon data (Sharpley & Williams, 1993). In this study, soil texture and organic carbon data provided by the ISRIC World Soil Information (Soilgrids data) were used to calculate the K factor by using the method of Wischmeier et al., 1971 (Eq. (3)). K defines both the vulnerability of soil erosion and the quantity and speed of runoff conditions. The erosive energy of rainfall has been quantified in terms of soil loss per unit area on a 22.1-m-long section of exposed soil with a slope of 9 % (Brady & Weil, 2012; Alexakis et al., 2013; Shit et al., 2015; Jiang et al., 2015). From the soil textural class, types, and organic matter, the K factor has been derived, calculated according to the United States Department of Agriculture (USDA) methodology (Shit et al., 2015; Biswas & Pani, 2015; Bhattacharya, Das Chatterjee, & Das, 2021).

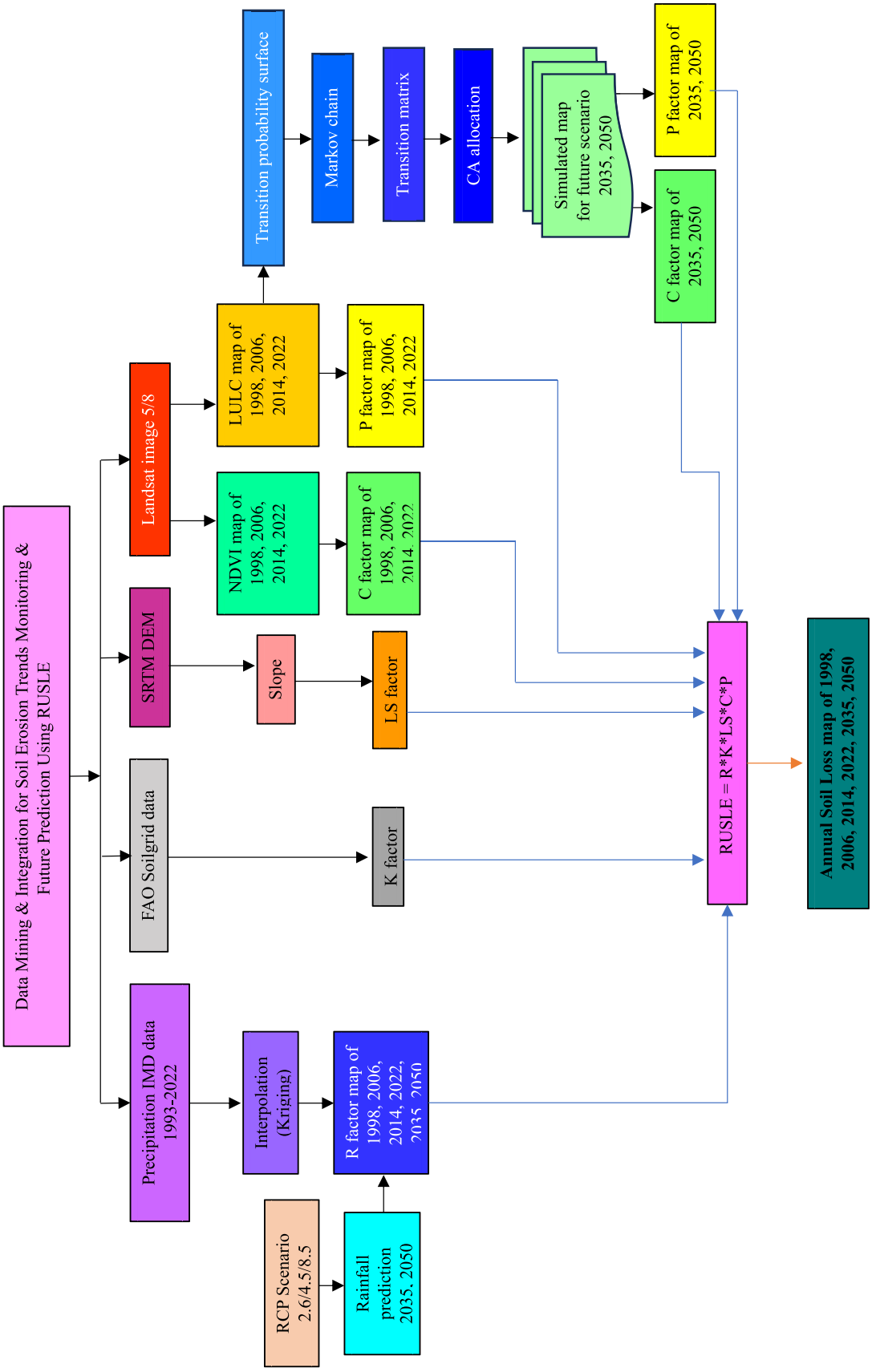


Fig. 2. Methodological flowchart of the study.

$$K = \left\{0.2 + 0.3\ exp\left[-0.0256 \times Sd \times \left(1 - \frac{St}{100}\right)\right]\right\} \times \left(\frac{St}{Cy + St}\right)^{0.3} \times \left[1 - \frac{0.25 \times OC}{OC + exp(3.72 - 2.95 \times OC)}\right] \times \left[1 - \frac{0.7 \times SM}{SM + exp(-5.51 + 22{\cdot}9 \times SM)}\right] \quad (3)$$

Where, SM = 1 − (Sd/100), Sd = Sand (%), St = Silt (%), Cy = Clay (%), OC = Organic carbon (%).

3.4. Topographic factor (LS)

The LS factor has been calculated using the Shuttle Radar Topography Mission (SRTM) DEM data of 30 m resolution. The L factor denotes the proportion of SE determined by the length of the field slope, where 72.6 feet is the standard for significant flow accumulation (Bhattacharya, Chatterjee, et al., 2021). The L factor is determined by applying Eqs. (4)–(6). The steepness of a slope (S) indicates the impact of the slope on erosion (Qiu et al., 2018; Tian et al., 2020). In order to overcome this limitation of the RUSLE calculation, which only considers slopes with a gradient of up to 18 %, was adjusted for steep slopes. This modification was proposed by Liu and Zhang in 2002 and further modified by Qiu et al. (2018).

$$L = \left(\frac{\lambda}{22.13}\right)^{m} \quad (4)$$

$$m = \begin{cases} 0.2 & \theta < 1\% \\ 0.3 & 1\% \le \theta < 3\% \\ 0.4 & 3\% \le \theta < 9\% \\ 05 & \theta \ge 9\% \end{cases} \quad (5)$$

Where the slope length factor is L; λ is the field slope length, and the exponent m varies with the slope θ. 22.13 m is a unit plot, and m is a changeable slope length exponent. Eq. (6) was used to figure out the S-factor.

$$S = \begin{cases} 10{\cdot}8\ sin\ \theta + 0.03 & \theta \le 9\% \\ 16{\cdot}8\ sin\ \theta - 0.05 & 9\% \le \theta < 18\% \\ 21{\cdot}9\ sin\ \theta - 0.96 & \theta \ge 18\% \end{cases} \quad (6)$$

The slope steepness factor (S) is derived from the slope angle θ. The Raster Calculator tool was used to determine the L and S factors in ArcGIS 10.8.2 software. The L factor and S factor were then multiplied to create the LS factor map.

3.5. Crop management factor (C)

The C factor plays an eventual role in erosion, as its a manageable circumstance that can help mitigate soil loss (Chatterjee et al., 2014; Renard et al., 2011, pp. 137–167). The presence of more vegetation cover leads to a significant reduction in soil erosion (Shit et al., 2015; Jiang et al., 2015; Hou et al., 2020). Tree cover mitigates erosion by capturing rainfall, improving infiltration, and decelerating runoff (Gyssels, Poesen, Bochet, & Li, 2005; Wang et al., 2013). The NDVI-based approach is the most frequently employed method for estimating the C factor; however, other methods are also available (Mirakhorlo & Rahimzadegan, 2020; Jaman et al., 2024). The C factor in this study was determined using the NDVI method described by Van de Knijff et al. (2000) in Eq. (7).

$$C = exp\left[-a\frac{NDVI}{(b - NDVI)}\right] \quad (7)$$

Where a and b are dimensionless parameters that govern the curvature of the concerning NDVI and C-factor.

Landsat TM 5 and 8/9 (OLI) images were utilized to generate NDVI maps provided by the USGS, and subsequently, a C factor thematic layer map was created. For calculating the C factor, the parameters α and b were set to 2 and 1, respectively, as per Van der Knijff et al. (2000). These parameters established an exponential relationship between NDVI and the C factor. A spatially distributed C factor map of the basin was produced by overlaying pixel-wise NDVI values on the basin raster map. The map takes into account the impact of crop types and reclamation practices on cropland and gullied areas to create an accurate representation of the C factor.

A traditional approach has been employed for future prediction of the C factor, where the C factor involves assigning values of LULC classes of the study region, as outlined in Table 4 (Mirakhorlo & Rahimzadegan, 2020). The LULC map has been prepared using Landsat data provided by the USGS for the last 24 years, and the future prediction has been made using the LCM (see Table 2).

3.6. Conservation practice factor (P)

The P factor is delineated as the ratio of soil loss resulting from a particular support practice to the resembling soil loss from a specific cropland, which has been calculated using an LULC map prepared from the Landsat image provided by the USGS. The future predicted LULC has been generated using LCM based on the last 24 years of LULC maps. The p-values presented in Table 3 are based on the standard table that was proposed by the USDA-SCS (1981) and Rao (2023). More effective conservation methods are indicated by the lower *p* values.

Table 3
Erosion rate.

Category	Erosion
very low	< 1 t ha^{-1} yr^{-1}
low	1–2 t ha^{-1} yr^{-1}
medium	2–5 t ha^{-1} yr^{-1}
high	5–10 t ha^{-1} yr^{-1}
very high	10–20 t/ha/year
extreme	> 20 t ha^{-1} yr^{-1}

Table 4
Land cover classification of the C factors.

S. no.	Land use/land cover classes	C value
1	Water bodies	0
2	Marshy land	0
3	Forest	0.5
4	Agricultural land	0.6
5	Built- up	0.02
6	Sand deposits	1
7	Fallow land	1

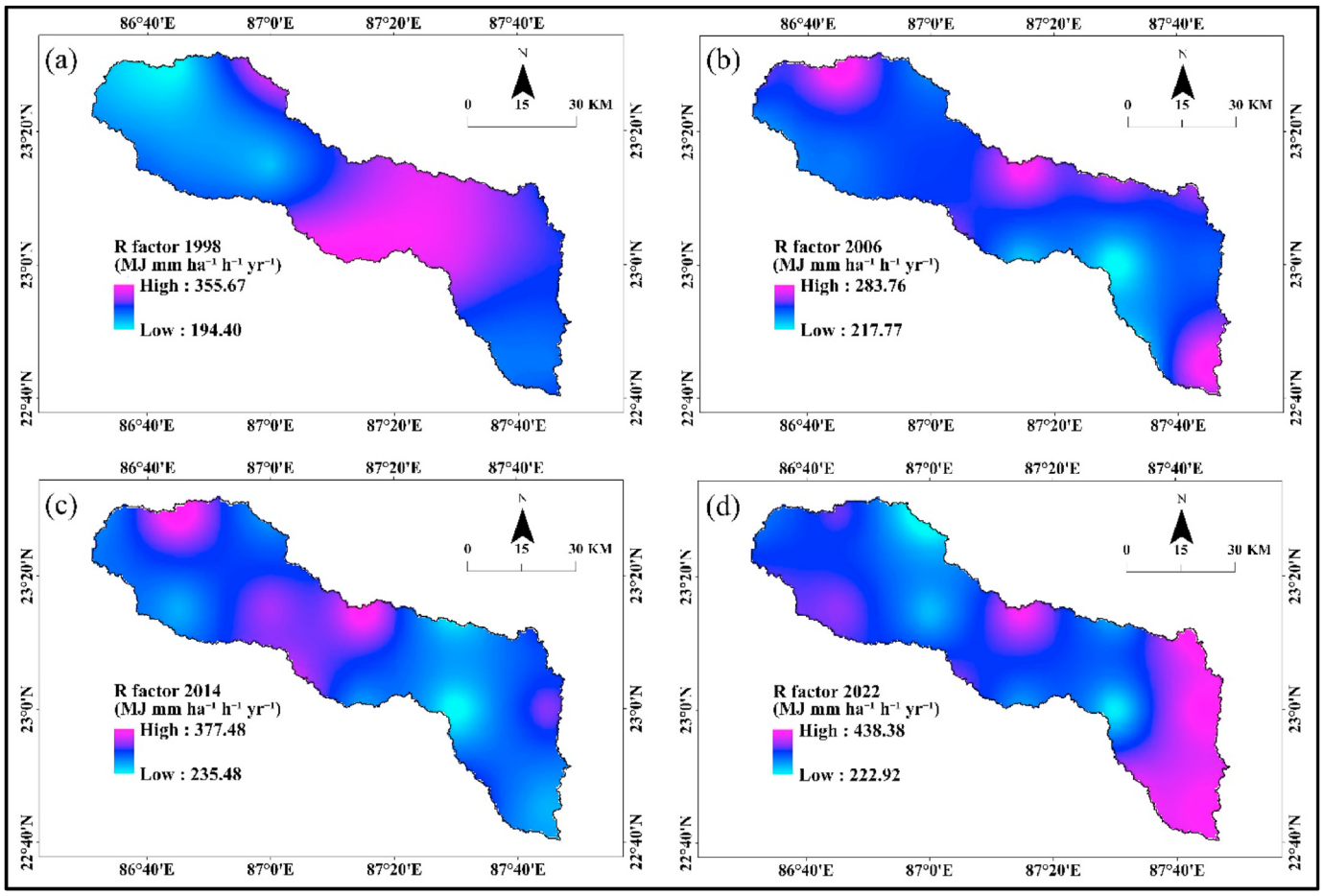


Fig. 3. Rainfall erosivity (R factor) maps of 1998(a), 2006(b), 2014(c), 2022(d).

3.7. Simulated land use land cover (LULC) map for future

To simulate future land use changes, the LCM is a powerful tool for comparison with baseline periods (Anand et al., 2018; Chuenchum et al., 2020b; Ait El Haj et al., 2023). LCM is a module designed for land planning and decision support aiming at ecological sustainability, and is fully integrated within the TerrSet program developed by Clark Labs at University of Clark (Nor et al., 2017; Ait El Haj et al., 2023). According to Perez-Vega et al., (2012), LCM can evaluate current and future land use and land cover changes based on empirical model relationships with explanatory variables such as slope, elevation, distance to roads, distance to rivers, and distance to city centres, among others, to simulate future scenarios. The simulation process in the LCM module is divided into three stages: change analysis, transition potentials, and change prediction. These methods rely on historical land change data from specific periods (Table 4) and projected future land scenarios (Negese et al., 2022; AitEl Haj et al., 2023).

To study land use and land cover monitoring and forecasting, one must understand the LCM model simulation of TerrSet software. The first stage involves change analysis, comparing land cover maps from baseline periods, which are influenced by factors such as urbanization, population growth, and natural resource demand. The second stage utilizes transition potentials, generated by a multi-layer perceptron (MLP) artificial neural network to create transition potential maps (Rumelhart et al., 1986; Negese et al., 2022). These maps are refined by reducing the RMSE error to enhance accuracy. Factors such as elevation, slope, proximity to roads, proximity to rivers, distance from urban centres, and others affect simulated land use changes. The LCM module concludes with change prediction, using historical change rates and transition potential models to estimate future scenarios. Markov chain analysis estimates the relative amount of future land use transition (Arumugam et al., 2021; Gemmechis, 2022; Norris, 1997).

The projected future land use map is then calibrated and validated using baseline maps, employing three Kappa indices for accuracy assessment (Arumugam et al., 2021): the traditional Kappa index (Kstandard), Kappa for no ability (Kno), and Kappa for quantity and location (Klocation & Klocation stratum). These indices consider varying accuracy levels and indicate the consistency between two categorical datasets and class size distributions (van Vliet et al., 2011; Chuenchum et al., 2020b). Eqs. (8)–(10) show the Kstandard, Kno, and Klocation & Klocation strata:

$$K_{standard} = \frac{P_0 - P_e}{1 - P_e} \tag{8}$$

$$K_{no} = \frac{P_{max} - P_e}{1 - P_e} \tag{9}$$

$$K_{location} = \frac{P_0 - P_e}{P_{max} - P_e} \tag{10}$$

Where P_0 is the observed agreement fraction, P_e is the expected agreement fraction, and P_{max} is the maximum agreement fraction. The Kappa index ranges from 0 to 1, with lower values indicating

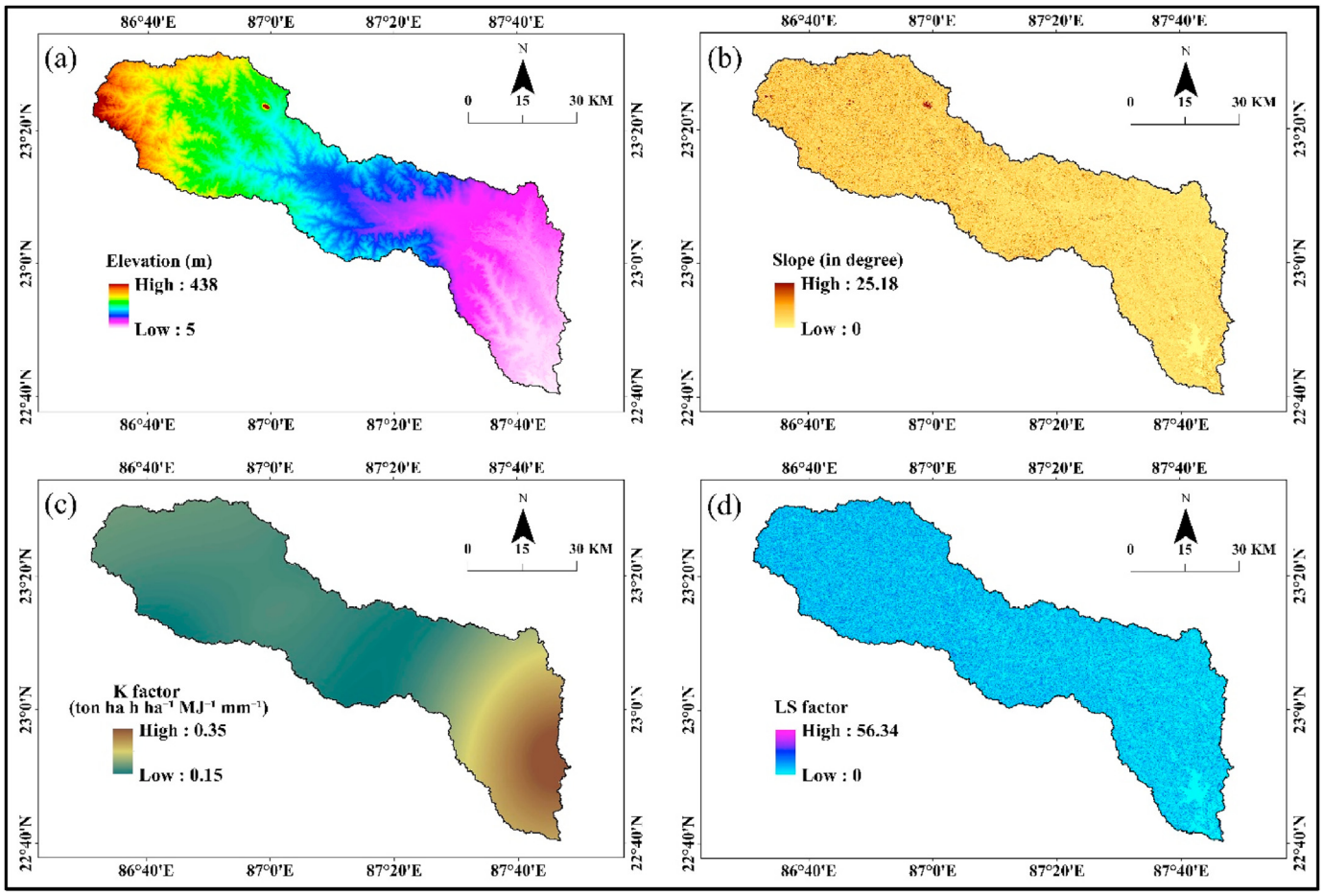


Fig. 4. (a) Elevation map, (b), slope map (c), soil erodibility (K factor) map (d), Slope length and steepness (LS factor) map.

less matching. Consistent Kappa index values mean the simulated land use map is reliable.

4. Results and discussion

4.1. RUSLE model parameter calculation

4.1.1. Climate change scenario (R factor)

This study observed the monthly average rainfall of 42 IMD grid points from 1993 to 2022 (Fig. S1), to estimate the R factor. The spatial variation of the R factor has been computed using Eq. (2), which is shown in Fig. 3, indicating a direct linear relationship between rainfall erosivity and rainfall depth. Using these values, the inverse distance weighting (IDW) approach was used to create a spatial distribution map of the R factor. From 1993 to 2022, the average monthly rainfall was approximately 113.08 mm, with a maximum of 338.86 mm and a minimum of 12.59 mm. The *R* factor values for the year 1998 ranges from 194.41 to 355.67 MJ mm ha^{-1} h^{-1} yr^{-1}, in 2006 ranges from 217.77 to 293.92 MJ mm ha^{-1} h^{-1} yr^{-1}, in 2014 ranges from 235.48 to 377.84 MJ mm ha^{-1} h^{-1} yr^{-1}, and the values for 2022 ranges from 222.93 to 438.39 MJ mm ha^{-1} h^{-1} yr^{-1} respectively (Fig. 3). The mean future R values under RCP scenarios (2.6, 4.5, 8.5) in 2035 were 23.56 MJ mm ha^{-1} h^{-1} yr^{-1}, 22.96 MJ mm ha^{-1} h^{-1} yr^{-1}, and 22.84 MJ mm ha^{-1} h^{-1} yr^{-1}. and 25.16 MJ mm ha^{-1} h^{-1} yr^{-1}, 24.76 MJ mm ha^{-1} h^{-1} yr^{-1}, and 25.52 MJ mm ha^{-1} h^{-1} yr^{-1}, in 2050 (Figs. 9(a–c) and 10(a–c)).

Future scenario R factor maps indicate that rainfall erosivity (R factor) is expected to decrease in many regions due to the impacts of global warming and climate change. The same condition has been explained by Datta and Das (2019): "The Sub-Himalayan region and the western parts of West Bengal (WB)" will experience a declining trend of rainfall. According to Pal et al. (2015) and Mukherjee (2017), WB indicates a decreasing trend in monsoon rainfall. This decline is primarily linked to a projected change in various climatic issues to cause SE. Since the R factor is closely correlated with the intensity and amount of precipitation, its spatial distribution tends to follow a similar pattern to that of total precipitation. Figs. S1 and S5 compare the average annual historical rainfall with projected rainfalls in 2035 and 2050 under various RCP scenarios in the DRB. The projections show similar patterns but differ in magnitude and peak rainfall for some months, affecting drought and flood events. The number of projected rainfall events is lower than historical rainfall, with RCP4.5 scenarios generally underestimating magnitude. A similar result has also been found in Tuvshinjargal and others' work (2024) in Mongolia, and another study by Teng et al. (2018) predicted the R factor using different climatic models, i.e., PSL-CM5A-LR, GFDL-CM3, and MIROC-ESM, which showed a decreasing trend. Eq. (2) was used to compute the future R factor for 2035 and 2050 under RCP scenarios (Behera et al., 2020; Hateffard et al., 2021; Jaman et al., 2025; Sathiyamurthi et al., 2023). The R factor ranges were 214–237, 208–232.52, 207–231.32 MJ mm ha^{-1} h^{-1} yr^{-1} for RCP 2.6, 4.5, 8.5 in 2035. In 2050R factor ranges were 221–248.57, 219–241.34, 225–250.35 MJ mm ha^{-1} h^{-1} yr^{-1} for RCP 2.6, 4.5, 8.5. The spatial distribution of the R factor in 2035 and 2050 exhibits similar patterns but varying magnitudes due to global warming and climate change.

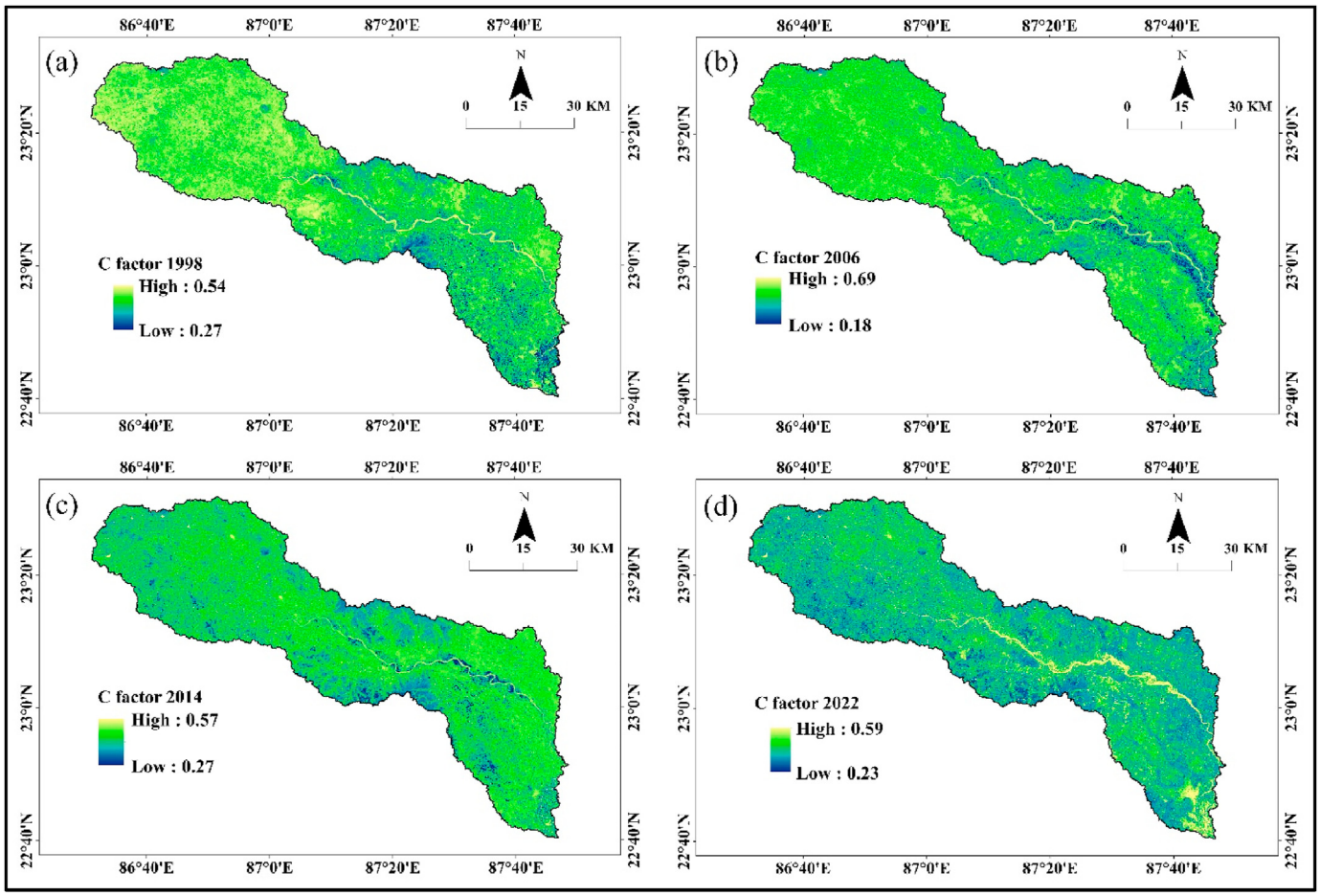


Fig. 5. Cover management (C factor) maps of 1998(a), 2006(b), 2014(c) and 2022(d).

4.1.2. K factor

The soil erodibility (K) factor ranges from 0.153 to 0.359 ton ha h ha^{-1} MJ^{-1} mm^{-1}, with a mean value of 0.20 ton ha h ha^{-1} MJ^{-1} mm^{-1} (Fig. 4(c)). The higher values are in the southeastern area, and the lower values are spread across the basin. In the study area, silt, clay, and organic carbon contents show consistent patterns across the basin, while sand content is more pronounced in the middle. Soil organic carbon ranges from 1.42 % to 3.14 %, clay from 17.1 % to 37.7 %, silt from 20.6 % to 51.4 %, and sand from 19.5 % to 63.2 %. The K factor values are lower in the upper and middle portions of the basin, with higher erodibility in cultivated areas and lower erodibility in high-relief areas. Specifically, the southeastern regions of the DRB exhibit the high K values, which is attributed to the high clay content coupled with lower organic matter, hindering the formation of protective clay-humus complexes. Consequently, the soil permeability in this domain remains relatively low, resulting in a high runoff potential (de Faria Godoi et al., 2021; Jaman et al., 2025; Roy and Sengupta, 2023; Tuvshinjargal et al., 2024; Zafor et al., 2023). This makes the lower basin susceptible to particle detachment through splash erosion and surface runoff.

4.1.3. LS factor

The L (slope length) and S (slope steepness) factors were calculated independently and then combined to derive the LS factor, which represents the topographic influence on soil erosion. In the study area, LS factor values range from 0 in flat, low-lying regions to a maximum of 56.34 in high elevation zones, predominantly located in the upper part of the basin (Fig. 4(d)). The mean LS factor across this basin is 0.81. Low values correspond to gently sloping or flat areas, whereas steep and elongated slopes in high elevated regions exhibit significantly higher LS values relative to the other regions. These elevated LS factor values in steep slope areas enhance the potential for soil erosion, emphasizing the strong influence of topography on erosion risk in the DRB. The entire basin is characterized by a very low LS value, with some isolated areas having the highest LS value such as, Susuniya Hill and the upper part of the basin area.

4.1.4. C factor

Low C factor values correspond to dense evergreen forests, while high C factor values indicate bare rocky terrains (Fig. 5). The C factor value ranges from 0.27 to 0.54 in the year of 1998 with a mean value of 0.43; the C factor value ranges from 0.18 to 0.69 in the year of 2006 with a mean value of 0.37; the C factor value ranges from 0.27 to 0.57 in the year of 2014 with a mean value of 0.33; and the C factor value rages from 0.23 to 0.59 in the year of 2022 with mean value of 0.38.

The 80 % areas of DRB have higher C values. In contrast, the smallest C value is found in forest areas in the middle part of the basin near Bishnupur, Joypur, Bhara, Birsingha, Jangal Mahal and Kodalia. Some large, patchy vegetation is present in the upper part of the basin near Susunia, Kapasberia, Pokariatol, and other areas. The upper areas of the DRB, mainly in the Prulia district, are covered by shrubs, grass, and hard rock bare land with a high C value. In future, the C value will be increased further according to the future predicted map (Fig. 8(c) and (d)).

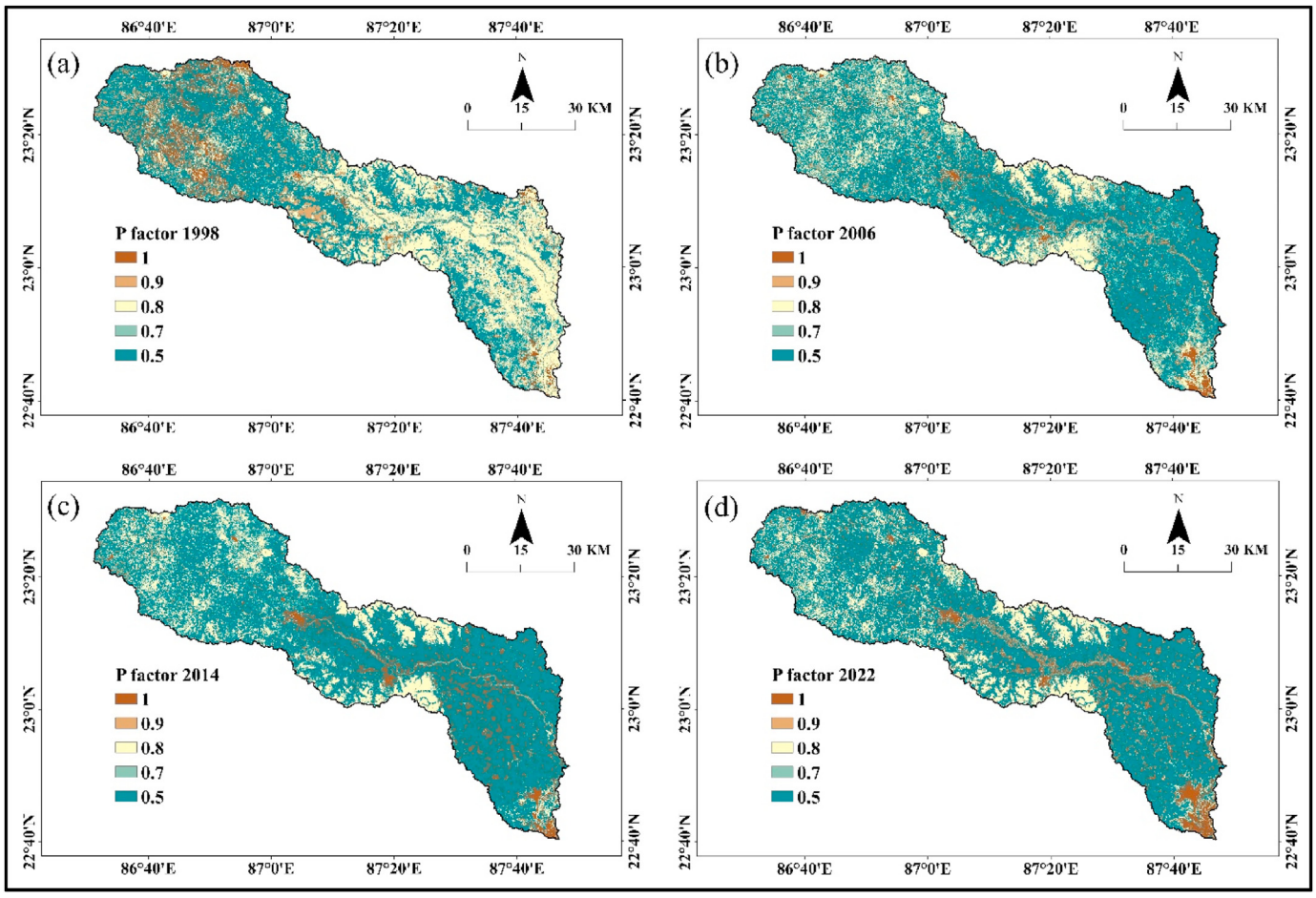


Fig. 6. Support practice (P factor) maps of 1998(a), 2006(b), 2014(c), 2022(d).

Table 5
Land cover classification of the P factors.

S. no.	Land use/land cover classes	*p* value
1	Water bodies	1
2	Marshy land	1
3	Forest	0.8
4	Agricultural land	0.5
5	Built- up	1
6	Sand deposits	0.7
7	Fallow land	0.9

4.1.5. *P factor*

The P factor values ranged from 0.5 to 1, with the highest values found in water bodies, wetlands, built-up areas, and sand deposit areas (Fig. 6(a–d)), assuming that no conservation works have been carried out. The forest areas were characterized by moderate *p* values, whereas the *p* values were low in agricultural land. Most of the areas of DRB fall under agricultural land, where 50 % of the areas were covered with agricultural land in 1998. It increased to 80 % of areas, now covered with agricultural land. In the future, this area is expected to increase further according to the predicted map (Figs. 9(d) and 10(d)).

4.1.6. *Simulation of land use change*

This study analyzed the evolution of LULC and transition potential over time, i.e., 1998, 2006, 2014, 2022, 2035, and 2050 (Figs. 2 and 8(a), (b)). The Kappa coefficients for the years under investigation were approximately 0.91, 0.89, 0.92, 0.93, 0.89, and 0.87, respectively, which suggests that these values were substantial. Overall, the land use maps in 1998, 2006, 2014, 2022, 2035, and 2050 had accuracies of 95.8 %, 92.9 %, 96.3 %, 97.2 %, 92.8 %, and 90.2 %, respectively, based on a general accuracy assessment comparing observed and simulated maps.

The LCM module utilized an MLP artificial neural network (ANN) to simulate transition potential maps, illustrating the shift from one land use to another. These maps were based on six sub-models, each corresponding to one of the following explanatory variables: elevation, slope, road network, distance from the river, distance from the road, and distance from the urban center. Tables 5 and 6 present the results of the assessment of the MLP ANN in the context of transition potential modeling, showing accuracy rate, training RMSE, and testing RMSE values. The accuracy and reliability of the six sub-models ranged between 66.5 % and 83.4 %. This demonstrates that using the MLP ANN within the LCM module is effective for simulating land use changes.

4.1.7. *Land use projection using LCM*

The prediction of land use change was based on transition probability, which represent the likelihood of land use transitions from 1998 to 2035, utilizing the Markov chain method. The highest transition probability was from natural vegetation to agricultural land and from agricultural land to built-up areas, while the lowest was from fallow land to built-up areas (Table 6). The land use map simulations for 2035 and 2050 were derived from a transition probability matrix covering the years 1998–2035 and 2050.

Table 6
Area in km^2 and percentage share.

Year	1998		2006		2014		2022		2035		2050	
Class Name	Area in sqkm	Area in percentage	Area in sqkm	Area in percentage	Area in sqkm	Area in percentage	Area in sqkm	Area in percentage	Area in sqkm	Area in percentage	Area in sqkm	Area in percentage
Waterbody	68.2236	1.57165791	53.4348	1.2309703	97.9335	2.2560809	80.4708	1.853795	102.695	2.3657709	116.6060	2.6862368
Marshy Land	394.4412	9.08668895	212.3658	4.8922424	140.0652	3.2266632	215.3736	4.9615327	57.549	1.3257486	52.9810	1.2205162
Natural Vegetation	1363.2903	31.4059355	1090.3005	25.117106	564.1596	12.996469	802.5174	18.487486	688.166	15.853188	653.8820	15.063392
Agricultural Land	2011.9068	46.3480267	2737.5669	63.064961	3216.6774	74.102165	2839.653	65.416705	3006.783	69.266856	2988.2700	68.840375
Built-up Area	32.7753	0.75504018	66.6450	1.5352919	144.9972	3.340281	243.468	5.6087396	326.906	7.5308896	355.5480	8.1907115
Sand Deposit	49.6053	1.14275063	26.6337	0.613557	55.5921	1.2806677	45.4509	1.0470463	50.487	1.1630691	65.9293	1.5188044
Fallow Land	420.6258	9.68990006	153.9216	3.5458712	121.4433	2.797673	113.9346	2.6246961	108.282	2.4944779	107.6520	2.4799647

For the year 2035, the land use map calibration included accuracy assessments using four Kappa statistics: $K_{standard}$, K_{no}, $K_{location}$, and $K_{location\ strata}$ (Chuenchum et al., 2020a; AzimiSardari et al., 2019; van Vliet et al., 2011). The values for $K_{standard}$, K_{no}, $K_{location}$, and $K_{location\ strata}$ were 0.948, 0.877, and 0.963, respectively. These Kappa statistics indicate that the LCM module in the TerrSet software effectively predicts LULC changes for 2035, and 2050 utilizing the Markov chain approach (Fig. S4).

4.1.8. Trends of land use change using LCM

The study used the LCM module (TerrSet software), which found notable changes in land usage between 1998 and 2035 and 2050. Most of the areas have changed, with natural vegetation transforming into agricultural land and agricultural land becoming built-up areas (Fig. 8a, b & Tables 7 and 8). Marshy and fallow lands saw declines of 85.41 % and 86.57 %, and 74.25 % and 74.41 %, by 2035 and 2050, respectively. There was a decline in natural vegetation of roughly 49.12 % and 52.04 % (688.17 km2 and 653.88 km^2). There were increases in agricultural land of 49.45 % (3006.78 km^2) and 48.53 % (2988.27 km^2). Socioeconomic development was the driving force behind the urban area expansion, with increases of 8.97 % (326.90 km2) and 9.85 % (355.55 km^2). There was little change in the sand deposits and water bodies.

The study found that the DRB region faces challenges like deforestation, land degradation, and population growth, leading to the conversion of agricultural land into built-up areas. This land use land cover change impacts the C factor and P factor, which are crucial for soil protection against erosion. The destruction of forest areas increases C factor values and decreases P factor values, resulting in higher soil erosion rates and potential extreme erosion in specific areas.

4.2. RUSLE model simulation

The RUSLE is a straightforward model that utilizes real-world data to predict the long-term average annual SE rate. It considers variables such as rainfall, soil type, topography, crop systems, and management practices. R, K, LS, C, and P in the GIS environment are used to predict the amount of soil that could be lost each year. The SE of 1998, 2006, 2014, and 2022 were used as a base period for simulating the future SE. Soil loss varied from 0 to 216.80 t ha^{-1} yr^{-1} in 1998. Soil loss varied from 0 to 443.53 t ha^{-1} yr^{-1} in 2006. Soil loss varied from 0 to 480 t ha^{-1} yr^{-1} in 2014. Soil loss varied from 0 to 423.28 t ha^{-1} yr^{-1} in 2022. It is observed that the areas with steep slopes and little vegetation were more likely to have SE (Fig. 7). Dense vegetation helps prevent SE even on steep hills, although low-vegetation areas washed away even when the slope was modest. SE is triggered by both slope and vegetation, but vegetation was more important. These results also showed that the location and height of plants had a good impact on soil erosion. The weather, topography, and vegetation were the primary factors influencing the SE rate.

Projected SE are made for two time periods: 2035 and 2050, which are soon to occur in near future (Figs. 11 and 12). In 2035, soil erosion rates varied from 0 to 780.34 t ha^{-1} yr^{-1} (RCP2.6), 0 to 657.71 t ha^{-1} yr^{-1} (RCP4.5), and 0 to 803.73 t ha^{-1} yr^{-1} (RCP8.5) (Fig. 11(a–c)). In 2050, rates varied from 0 to 770.71 t ha^{-1} yr^{-1} (RCP2.6), 0 to 744.07 t ha^{-1} yr^{-1} (RCP4.5), and 0 to 814.09 t ha^{-1} yr^{-1} (RCP8.5) (Fig. 12(a–c)).

All RCP model projections indicated that the majority of the DRB are expected to experience high SE rates in both 2035 and 2050, which is based on the influence of erosion agents. However, certain areas, particularly in the upper river basin and the central part of the basin, are projected to face a mild to high level of erosion. When compiling the soil erosion results from 1998 through to the future

Table 7
Probability of transition matrix from 1998 to 2035.

	water	wetland	forest	agriculture	buildup	sand	fallow
water	0.717	0.1341	0.0012	0.118	0.0147	0.013	0.002
wetland	0.1258	0.2502	0.0517	0.4201	0.1405	0.0054	0.0063
forest	0.0024	0.0353	0.4734	0.4231	0.0514	0.0018	0.0126
agriculture	0.0046	0.0002	0.0901	0.822	0.0564	0.0046	0.0221
buildup	0.0014	0.0003	0.0031	0	0.9137	0	0.0815
sand	0.1376	0.0224	0.0031	0.0751	0.0054	0.7253	0.0311
fallow	0.0059	0.0068	0.2336	0.5113	0.0065	0.0238	0.2121

Table 8
Probability of transition matrix from 1998 to 2050.

	water	wetland	forest	agriculture	buildup	sand	fallow
water	0.4615	0.1272	0.0012	0.1905	0.0317	0.1592	0.0287
wetland	0.1187	0.1542	0.0781	0.4873	0.1408	0.0017	0.0192
forest	0.0077	0.0144	0.5829	0.3131	0.0614	0.0036	0.0169
agriculture	0.0101	0.00046	0.1131	0.78854	0.0564	0.0068	0.0246
buildup	0.0001	0.0001	0.0004	0	0.9944	0.0001	0.0049
sand	0.1637	0.0358	0.0215	0.1418	0.011	0.5455	0.0807
fallow	0.0092	0.0235	0.2214	0.5904	0.0262	0.0248	0.1045

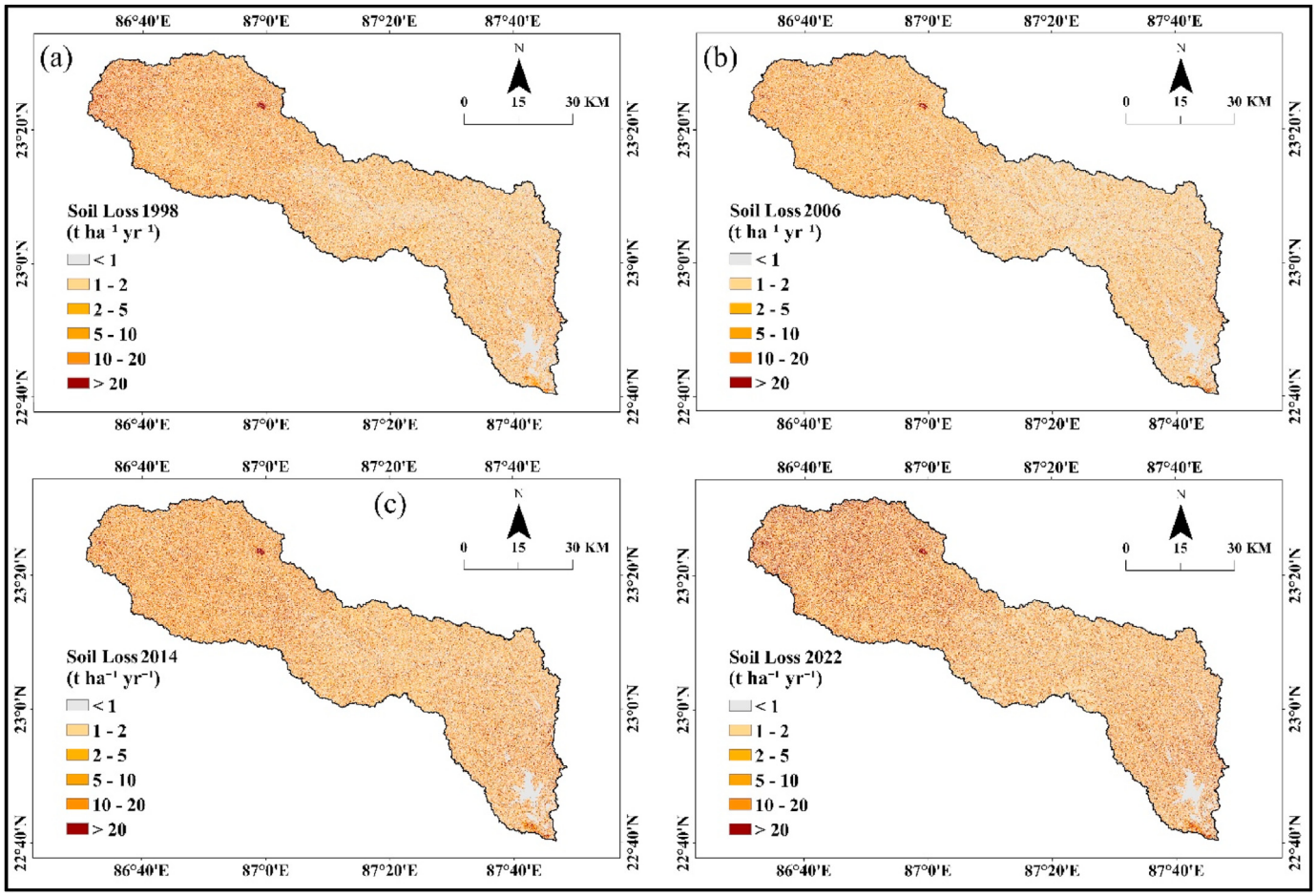


Fig. 7. Soil erosion maps of 1998(a), 2006(b), 2014(c), 2022(d).

projections, a trend emerges that closely mirrors the pattern observed in the R factor. Between 1998 and 2022, both the maximum and minimum erosion incidences increased, indicating a rise in the variability and intensity of erosion events. However, these projections suggest a reversal of this trend in the coming decades, with a gradual decline in erosion severity expected under future climate scenarios.

The distribution of erosion showed that steep hills and poor cover management led to high erosion rates in these areas. For each study year, the area for each grade and the portion of the area are

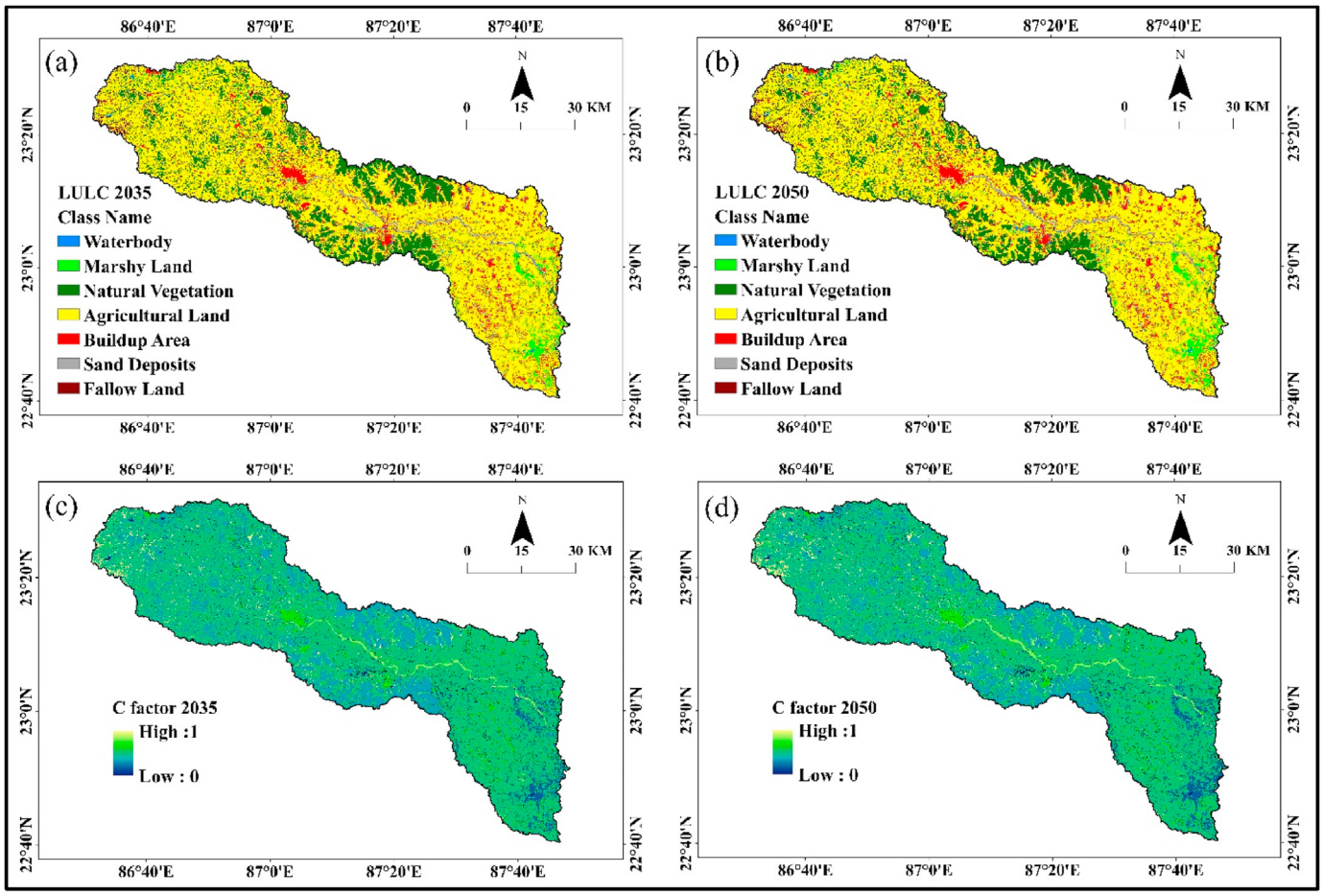


Fig. 8. Different variable maps of LULC (a, b), & C factor (c, d), in near future of 2035 and 2050.

shown in Table 9. As of 2022, 69.74 % of the area was considered to have a very low erosion risk. The other 12.21 %, 15.18 %, 2.63 %, 0.21 %, and 0.02 % of the area were considered to have low, medium, high, very high, and extreme erosion risks, in that order. The most dangerous levels of soil runoff were between 10 and 20 t ha^{-1} yr^{-1} and over 20 t ha^{-1} yr^{-1}. Examining the results from 1998, 2006, 2014, and 2022, it seems that very high levels of SE went up while extreme levels stayed the same. However, predictions for 2035 and 2050 showed that high-erosion risk areas would be more likely to experience erosion where the R factor increases. The landscape where SE rates are more likely to have steep slopes and little greenery. Based on these findings, hilly parts of the DRB, especially the upper part of the basin, which includes the Purulia and Susunia hills, have a high risk of SE.

To find a trend in soil erosion, simulation data from four different time periods were compared. Fig. 8 shows the changes in SE from 1998 to 2006, 2014, and 2022. Fig. 11 and Fig. 12indicating how SE will change in 2035 and 2050. Table 9 shows that the levels of very high and extreme SE rose in 2022 compared to the three previous years, 1998, 2006, and 2014. It also indicates that the levels of very high and extreme soil erosion are likely to worsen in the future compared to 2022. The climate change scenarios represent that in the future, the R value will decrease. On the other hand, the intensity of SE in the future is expected to be higher than in 2022. This means that the C factor and the P factor have a big effect on soil erosion.

The present and future land cover maps were further assessed in terms of the linkage between soil erosion and land cover type in more detail. Terrset software has been used for simulations of LULC maps for 2035 and 2050 to predict the C and P factors, which help determine the SE rates influenced by climate change and land use changes. Table 9 shows how the rates of soil erosion changed over time based on 2022 and comparisons to the future. The SE rates are expected to continue rising in the future; however, erosion in fallow lands and sand deposits is higher, but not in urban areas, wetlands, and places with extensive vegetation. According to the results of the future scenario, there will be more soil erosion in cropland in 2035 and 2050. This means that policy steps need to be taken based on the type of land cover.

5. Discussion

Several studies have used individual climate models to project future monsoon rainfall under different GHG emission scenarios. This forcing consistently indicates enhanced thermodynamic conditions conducive to greater overall monsoon precipitation (Meehl & Arblaster, 2003; Sugi & Yoshimura, 2004). Specifically, by the late 21st century, DAIRAKU and EMORI (2006) and Kumar et al. (2006) forecasted a substantial 20–30 % increase in rainfall over the western Himalayas. Conversely, studies of Datta and Das (2019), Pal et al. (2015), and Mukherjee (2017) predict a reduction in rainfall in the Sub-Himalayan region, as well as the central and western parts of WB in future, which is similar to the result of this study.

A key innovation of this study lies in its novel prediction of the R, C and P factors of DRB, a comprehensive approach that has not been concentrated on. That distinguishes it from prior research. SE by

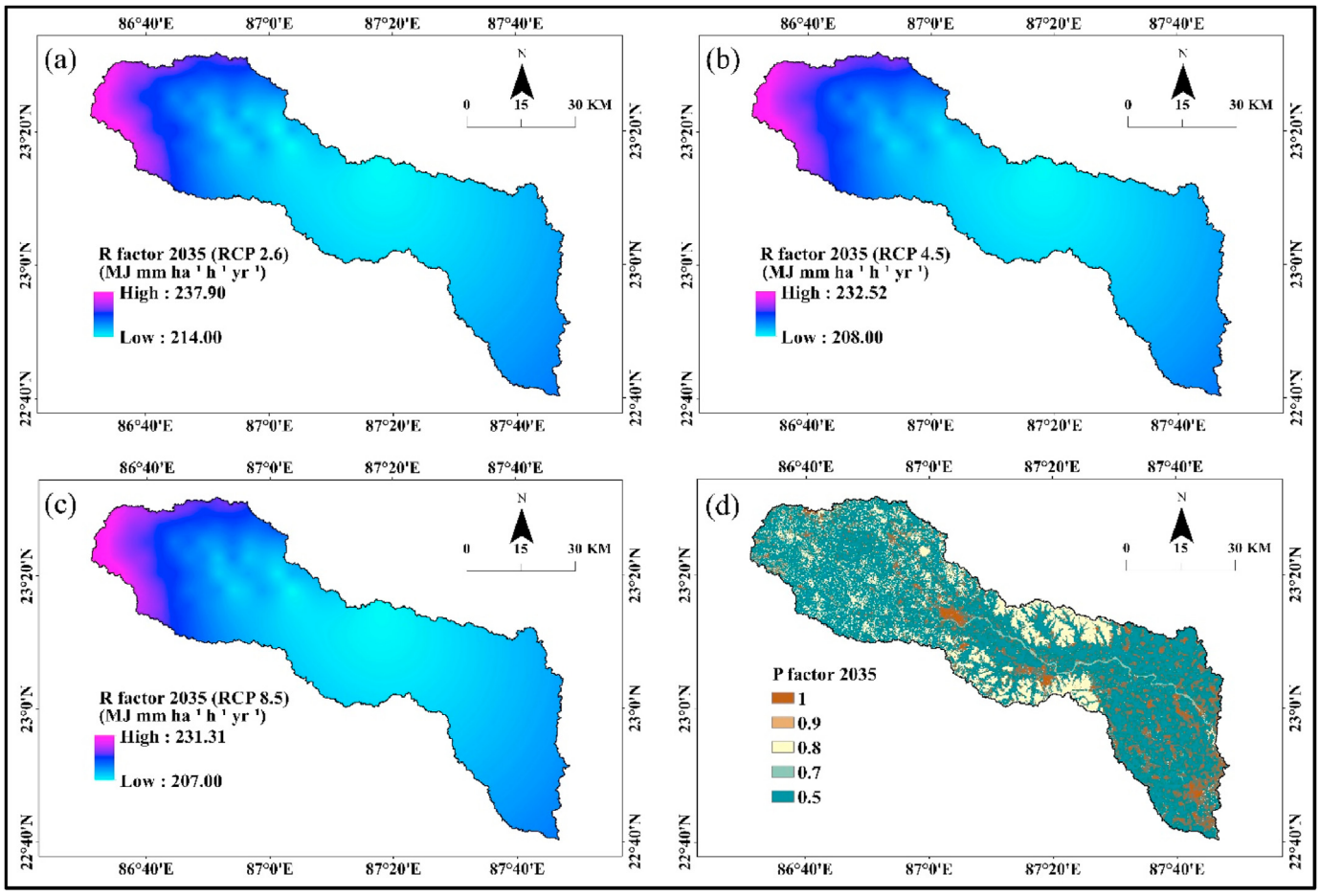


Fig. 9. (a), (b), (c) represents R factor maps of different RCP scenarios for near future 2035, and (d) represents the P factor map of 2035.

water poses a significant environmental challenge in the mixed climatic regions of WB, threatening cultivation and livelihood. While previous studies have assessed future SE in these areas based on future predicted R factor (Pal & Chakrabortty, 2019; Chakrabortty et al., 2020; Zafor, Kundu, & Maity, 2023).Investigations considering the combined influence of climate and land use and topography-related covariates on future SE rates across remote areas remain scarce. This study addresses this gap by examining the impact of future R, C, and P factors on SE in the semi-arid region of WB, highlighting potential land and biodiversity conservation. Although wind erosion was not considered here, future research should integrate both water and wind erosion assessments (Ekhtesasi & Sepehr, 2009).

Soil erosion values across the DRB basin increased from 1998 to 2014 and then showed a slight decline by 2022, which is mainly due to afforestation efforts and also activities in the few parts of the basin (Fig. S6); however, this study on assessing and predicting soil erosion is very crucial because the changing climate may worsen soil and land degradation (Wang et al., 2017), with consequent risks to biodiversity and ecology. Erosion levels in 2022 remained higher than in 1998, even as the R factor continued to rise in entire regions (Pal & Chakrabortty, 2019; Chakrabortty et al., 2020). Historical soil erosion show that extremely low and low erosion zones decrease in present time, while other classes rise. Projections suggest that changes in the R factor due to shifting rainfall patterns under climate change will likely reduce in 2035 compared to 2022, but it is expected to rise again by 2050 which is similar to the study of Tuvshinjargal et al. (2024). The upper parts of the DRB basin, characterized by minimal vegetation and steep slopes, are especially prone to severe erosion (Panagos, Borrelli, Poesen, et al., 2015; Garcia-Ruiz et al., 2015). The spatial distribution of soil erosion remains consistent with topographic features, as areas with higher LS factor values indicating longer and steeper slopes are more susceptible to erosion (Teng et al., 2018). These findings highlight that natural slope gradients play a critical role in determining erosion risk, with steeper terrains significantly increasing vulnerability (Lamchinet al., 2016).Precipitation variation is mainly driven by climatic change, which can significantly accelerate desertification and alter land cover, thereby influencing soil erosion dynamics. Rapid soil loss can occur due to the excessive rainfall, especially in semi-arid regions where vegetation is sparse, degraded, or unable to thrive due to prolonged dry conditions (Rivera-Marin et al., 2022; Zhang et al., 2019). Effective strategies such as restoring, expanding, and maintaining vegetation cover are essential in preventing soil erosion, as healthy plant cover can stabilize the soil erosion by reducing runoff.

LULC dynamics is another major driver of soil erosion (Borrelli et al., 2020; Cebecauer & Hofierka, 2008a, 2008b), as high vegetation cover reduces soil erosion. A comparison of LULC maps from 1998, 2006, 2014, and 2022 reveals a marked decline in fallow land, marshy land, and natural vegetation, while agricultural land has been increased rapidly over the years and it will increase in the future also (Table 6). This shift in land use has contributed to increased potential soil erosion (Zhang et al., 2019), particularly in fallow land areas, sand bodies, and agricultural tracts, where the protective vegetation cover is minimal or frequently disturbed

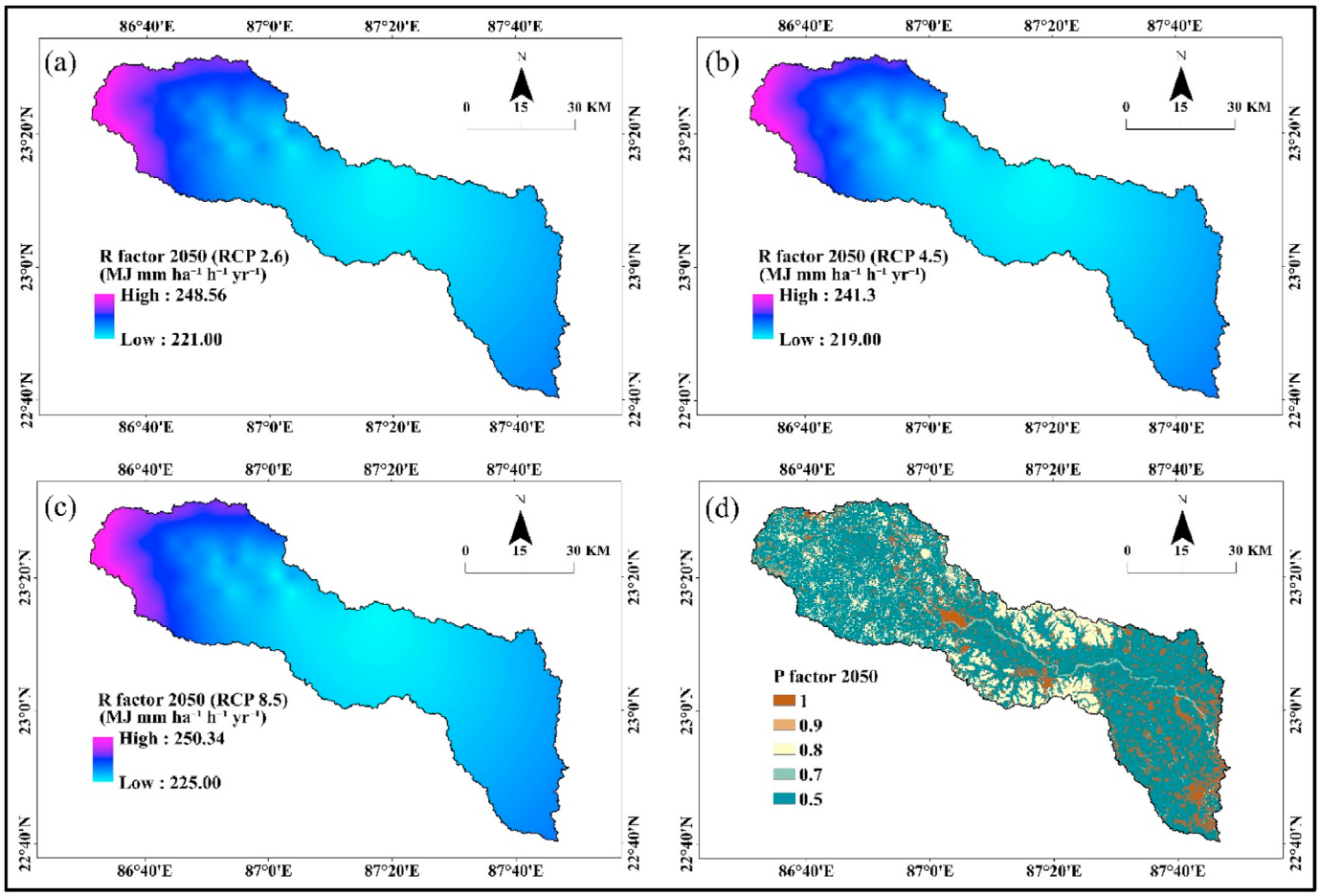


Fig. 10. (a), (b), (c) represents R factor maps of different RCP scenarios for near future 2050, and (d) represents the P factor map of 2050.

(Borrelli et al., 2020). Future projections of this region indicate that these trends will continue, leading to greater soil erosion in these vulnerable land cover types, underscoring the need for sustainable land management and conservation practices to mitigate further land degradation processes (Annabi, Le Bissonnais, Le Villio-Poitrenaud, & Houot, 2011; Zhang et al., 2019).

The annual crop management (C) factor is strongly affected by the intense spatial and temporal fluctuations in NDVI (Fig. S3) (de Carvalho Junior et al., 2014). The marked differences in C factor values between 1998 and 2022, 2006 and 2014 can likely be attributed to afforestation initiatives taken after 2014. Importantly, the C factor, like the Rainfall Erosivity (R) factor, exhibits significant sensitivity to semi-arid conditions and human impacts (Almagro et al., 2019). This study reveals that the combined influence of these two factors drives complex and accelerated SE. This study also influences the necessity of accounting for the changes in land uses, such as residential development, agricultural expansion, and bare land expansion, which assesses C factor variations (Karpilo Jr & Toy, 2004). In 1998 the mean value of C was 0.43, in 2006 it was 0.37, in 2014 it was 0.33, and in 2022 the value was 0.38. The C value increased slightly due to the afforestation that has been made in some pockets of the basin area. But in the future the higher value of the C factor area will increase due to increase of agricultural land according to the future predicted map (Fig. 8(c) and (d)).

On the other hand, LS factor andP factor were identified as the primary contributors to accelerated erosion. This observation is consistent with (Panagos, Borrelli, Poesen, et al., 2015; Comino et al., 2020), who emphasized the role of the P factor, estimating a potential 3 % reduction in soil erosion risk through practices like vegetation cover. However, the effectiveness of the P factor can vary with scale. The present study shows where 50 % of the areas were covered with agricultural land in 1998. It increased to 80 % of areas, now covered with agricultural land. In the future this area will be increased more according to the future predicted map (Figs. 9(d) and 10(d)).

These land cover changes play a pivotal role in erosion processes. To enhance the accuracy of future soil erosion predictions, both estimated C and P factors have been integrated into modeling approaches. This comprehensive method underscores the intricate link between land cover dynamics and soil erosion, offering a more efficient and contextually relevant foundation for future research and planning. This study recommends that policymakers and stakeholders focus on regions where both R, C, and P factors are projected to change dramatically. Such attention is vital to avoid irreparable losses in soil fertility and to manage the increased risk of floods and extreme sediment loads.

The DRB soil erosion evaluation and prediction showed that hilly regions with steep slopes and poor vegetation are more at risk. The upper part of DRB should be prioritized for nature conservation. Future soil management strategies can prioritize this area using this study. Human-caused soil disturbance has increased in DRB, while land-based research focuses on land use. According to the RUSLE simulation and soil erosion evaluation, increasing plant cover may safeguard soil and land resources. This will improve DRB soil health monitoring and environmental management. Climate change can be managed by national and international initiatives to

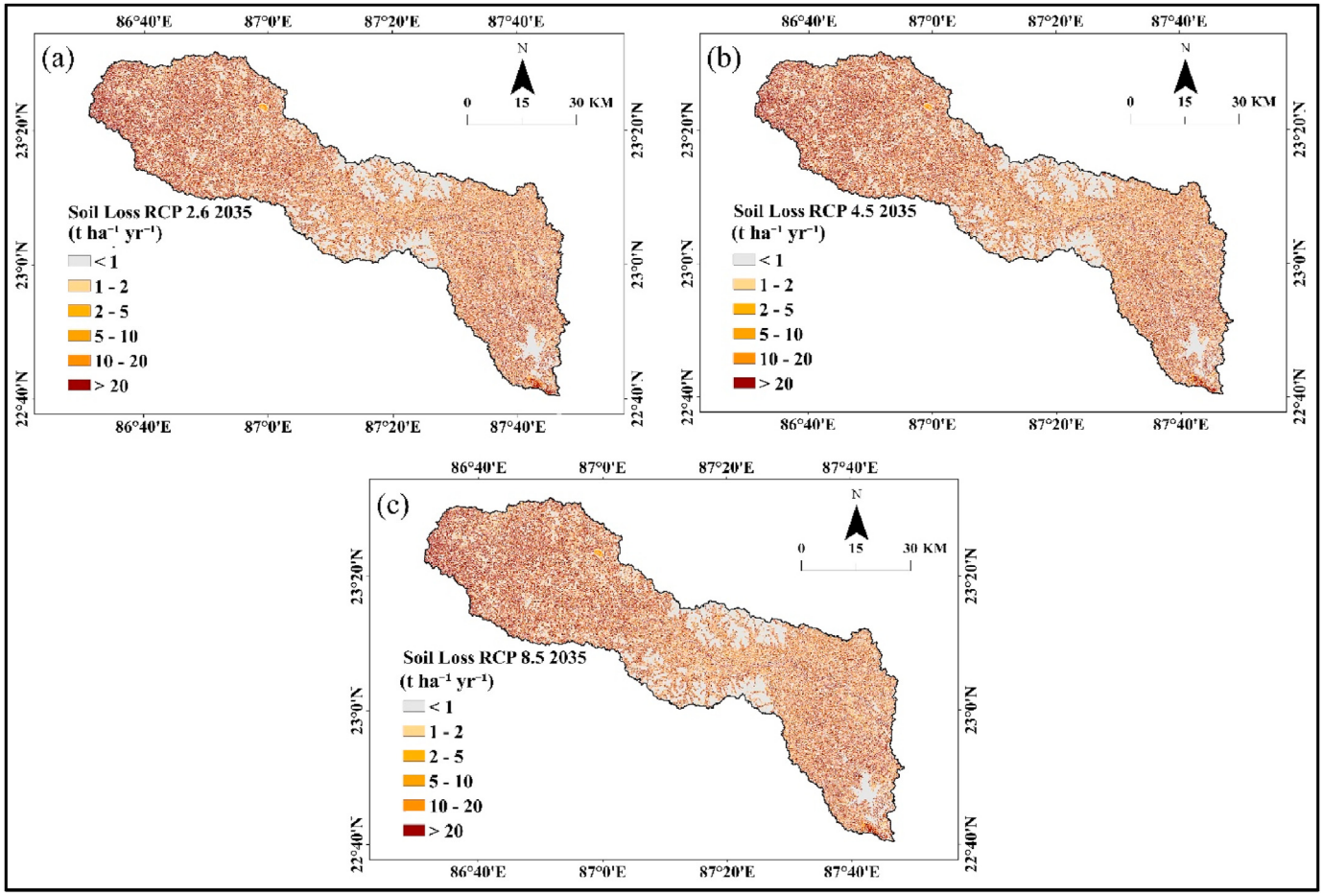


Fig. 11. (a), (b), (c) represents soil loss maps for near future (2035) under different RCP (2.6/4.5/8.5) scenario.

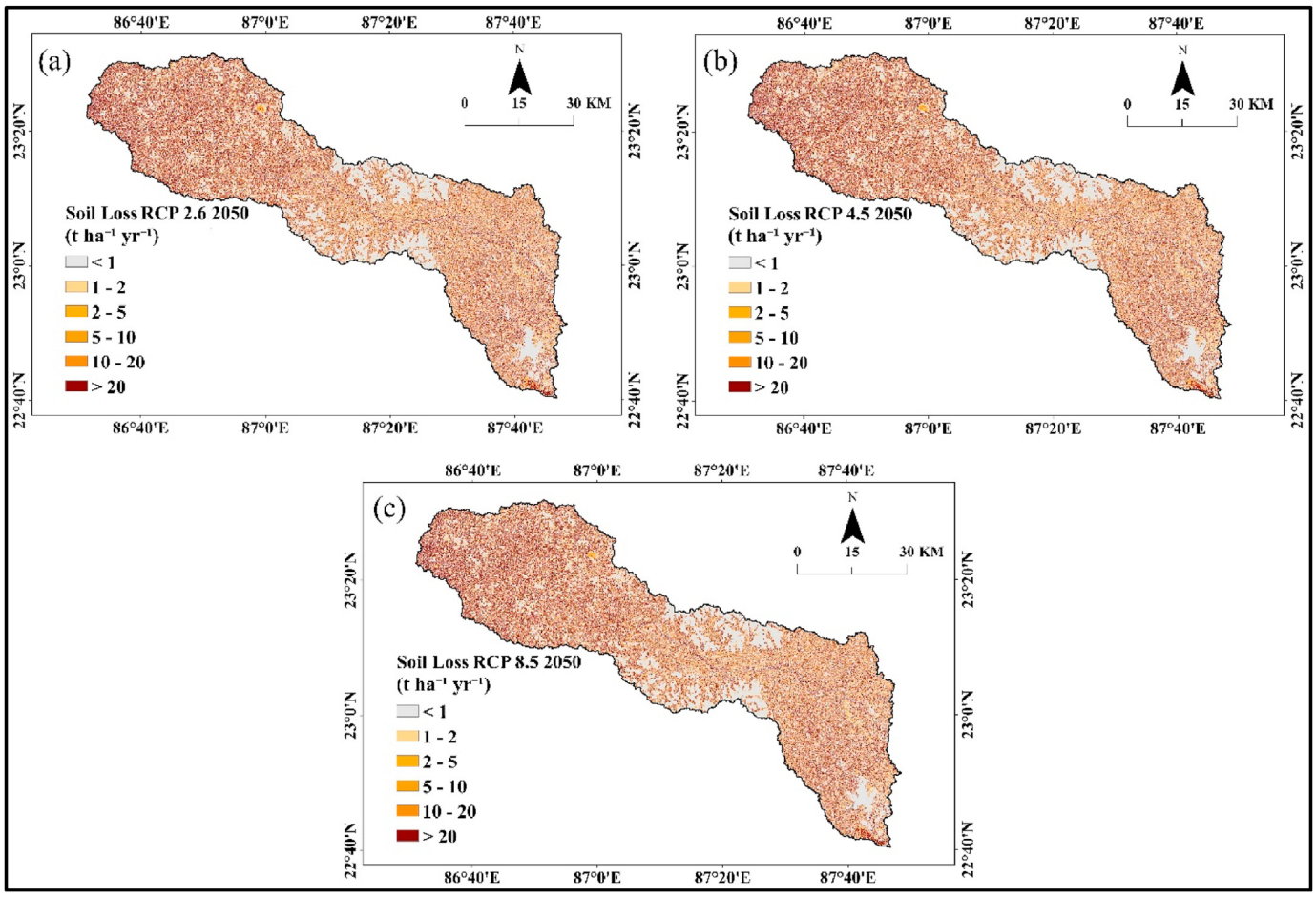


Fig. 12. (a), (b), (c) represents soil loss maps for far future (2050) under different RCP (2.6/4.5/8.5) scenario.

reduce greenhouse gas emissions; hence future changes should be predicted using several scenarios. This study offers a spatial view of future SE under different scenarios. While Uncertainties in inputs can affect the simulations. For example, the R factor, based on monthly/yearly precipitation averages across three RCPs, has inherent uncertainty, potentially explaining its weak correlation with the future. However, predicting future C and P factors accurately is also challenging due to its complexity and environmental sensitivity. In contrast, the spatial outputs for K and LS factors are reliable. Thus, this study provides significant and dependable spatial scenarios for future SE in the area.

6. Conclusion

The DRB annual SE risk assessments were analyzed using the RUSLE model. The model's simple equation reveals the region's most vulnerable locations for soil erosion and their spatio-temporal distribution. The regions high-resolution soil erosion maps have been estimated using baseline historical data from 1998, 2006, 2014, and 2022, and RCP 2.6, 4.5, and 8.5 scenarios simulated the near and far future soil erosion risk assessment. The upper part of DRB experienced the highest erosion rate due to its little vegetation cover and steep slopes. The middle of the basin had moderate erosion because of dense plant cover, while the lowest half of the DRB had rising erosion due to LULC change. Future scenarios revealed that socio-economic growth in this region forced most forests to become agricultural and urban areas, which boosted the C factor.

This study calculates soil erosion and identifies high-priority conservation areas to help improve soil erosion management. Most soil erosion occurs in fallow land, sand bodies, and agricultural land; thus, vegetation restoration should be a priority. DRB agricultural land dominates, covering over 60 % of the total area. With vegetation declining and deforestation worsening annually, regionally customized strategies are needed to preserve the ecological sustainability of the landscape (*p* factor).

This research is significant for its contribution to assure sustainable watershed and land resource management in a region vulnerable to both climatic and anthropogenic activities. By identifying current soil erosion zones and also projecting future vulnerable zones, this study offers valuable guidance for policymakers, planners, and local authorities in designing effective erosion control strategies. This research emphasizes the critical role of land use changes and climatic variability in shaping the nature and characteristics of soil erosion, making the study particularly relevant for policy development under changing environmental conditions. Furthermore, the methodology adopted in this study is adaptable to other river basins too, enhancing its utility for broader geospatial and environmental planning applications. The research thus bridges a critical gap in soil erosion modelling in the eastern India and promoting data-driven techniques, spatially targeted approaches to achieve environmental sustainability.

To enhance the model's accuracy, incorporating gully erosion, which is a significant contributor of SE in many regions, may increase the significance of the study. In the prediction of climate, no

Table 9
Soil erosion rates from 1998 to far future (2050) in different RCP scenarios (2.6/4.5/8.5).

Erosion class (t/ha/yr)		1998		2006		2014		2022		Near future period (2035)						Far future period (2050)					
										RCP 2.6		RCP 4.5		RCP 8.5		RCP 2.6		RCP 4.5		RCP 8.5	
		Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%	Area (km)	%
<1	Very low	3160.13	72.80	3557.27	81.95	2961.22	68.22	3027.19	69.74	2923.49	67.35	2925.19	67.39	2913.53	67.12	2900.51	66.82	2901.81	66.85	2899.34	66.79
1–2	Low	696.73	16.05	544.28	12.54	694.25	15.99	529.95	12.21	106.38	2.45	107.61	2.48	119.96	2.76	114.30	2.63	115.42	2.66	113.25	2.61
2–5	Medium	440.91	10.16	216.78	4.99	609.07	14.03	659.08	15.18	353.45	8.14	366.64	8.45	369.78	8.52	336.66	7.76	343.66	7.92	330.54	7.61
5–10	High	39.00	0.90	19.86	0.46	70.48	1.62	114.41	2.64	528.55	12.18	531.71	12.25	531.46	12.24	525.02	12.09	527.30	12.15	523.75	12.07
10–20	Very high	3.32	0.08	2.06	0.05	4.85	0.11	9.32	0.21	336.83	7.76	323.53	7.45	321.07	7.40	358.30	8.25	351.38	8.09	364.30	8.39
>20	Extreme	0.78	0.02	0.62	0.01	0.99	0.02	0.92	0.02	92.17	2.12	86.19	1.99	85.06	1.96	106.08	2.44	101.31	2.33	109.68	2.53

such single method is 100 percent reliable, so incorporating other models can justify the model accuracy.

CRediT authorship contribution statement

Basudeb Pyne: Writing – original draft, Validation, Investigation, Conceptualization. **Sushobhan Majumdar:** Formal analysis, Data curation. **Jaidul Islam:** Supervision, Methodology, Conceptualization. **Edris Alam:** Supervision, Software, Methodology, Funding acquisition. **Md Kamrul Islam:** Writing – review & editing, Funding acquisition, Data curation.

Declaration of competing interest

I, Dr. Jaidul Islam as the corresponding author of this article entitled as "**Analyzing Soil Erosion Trends and Future Predictions Using the RUSLE Model: A Case Study of the Dwarakeswar River, West Bengal**" would like to assure that this work described has not been published before; that it is not under consideration for publication anywhere else; that its publication has been approved by all co-authors. I also like to assure the there is no conflict of interest relate to the article and all the ethical standard have been maintained in the present work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.iswcr.2025.08.011.

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